

Biology, 2016 EC, natural science, entrance exam, round 1

Grade 12 · Biology · 2016 E.C.

Time allowed: 120 minutes

Questions: 92

Total marks: 92

Name _____ Grade _____ ID No. _____ Date _____

1. A transgenic organism is an organism that: (1)
 - A. can undergo chromosomal non-disjunction
 - B. replicates its DNA without control
 - C. has received genetic material of another organism
 - D. is artificially cultured and harvested
2. Which of the following explains how biology takes part in biosensor technology? Using microorganism to (1)
 - A. develop vaccines
 - B. monitor pollutants
 - C. extract metals from ores
 - D. produce antibiotics
3. In a textile industry, microorganism's commercial products play a significant role in removing stains like grease by changing: (1)
 - A. the temperature of insoluble large stain molecules
 - B. the acidic or basic nature of the stain molecules
 - C. the color of stain molecules to another color
 - D. large insoluble stain molecules into small and soluble ones
4. Which one of the following statements is correct about application of bio-pesticide in agriculture? It (1)
 - A. breakdowns glyphosate by an enzyme introduced into plant cell culture.
 - B. uses bacteria and fungi or their metabolites to kill insects.
 - C. produces insect resistant crops through modification of the plant genes using bacterial plasmids.
 - D. produces toxic chemicals by crops through modification of their genes.
5. Which one of the following statements correctly describes bacteria? (1)
 - A. They are single celled prokaryotic organisms.
 - B. They are multicellular eukaryotic organisms.
 - C. They are single celled eukaryotic organisms.
 - D. They are multicellular prokaryotic organisms.
6. Gene flow is the movement of genetic material (1)
 - A. that drives a population towards perfection.
 - B. without altering the allele frequencies.
 - C. without affecting the balance of gene pool of a population.
 - D. where organisms either join or leave a population.
7. Which one of the following options is a constituent of peripheral nervous system? (1)
 - A. Somatic sensory nerves, spinal cord and efferent nerves
 - B. Afferent nerves, brain and automatic motor nerves
 - C. Afferent nerves, efferent nerves and somatic motor nerves
 - D. Spinal cord, efferent nerves and somatic sensory nerves

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8. Which one of the following structures of neurons is correctly paired with its function? (1)
- A. Cell body - maintains structure and supply energy
 - B. Axon - receives signal from muscle
 - C. Dendrites - protects the neuron and insulate it
 - D. Myelin sheath - reduces nerve impulse transmission
-
9. Which one of the following statements explains how nerve impulse is occurred? (1)
- A. the movement of neurotransmitter molecules into myelin sheath.
 - B. charge difference across the membrane of axon.
 - C. balanced course across the membrane of dendrites.
 - D. the movement of neurotransmitter molecules out of cell body.
-
10. Which one the following statements explains how the allosteric inhibitors act on enzymes activities? By (1)
- A. binding to a site other than the active site and altering the shape of the active site
 - B. substrate and preventing it from binding to the active site
 - C. active site of an enzyme and blocking the substrate
 - D. active site and altering the shape of the enzyme
-
11. Why are enzymes considered special molecules in metabolism? Because (1)
- A. enzymes facilitate metabolic reactions under controlled physiological conditions.
 - B. metabolic processes can produce special products in the presence of enzymes.
 - C. enzymes are protein molecules that act as a source of energy.
 - D. reactions can't totally take place in the absence of enzymes.
-
12. Which of the following is the most advanced and recent event in the history of cell biology? (1)
- A. Enovation of the first scanning electron microscope
 - B. Discovery of the double-helix structure of DNA
 - C. Description of the Golgi apparatus
 - D. Sequencing of the human genome
-
13. Which one of the following properties permits insects to freely walk on water? Its high (1)
- | | |
|--------------------------------|---------------------------|
| A. adhesive force | B. specific heat capacity |
| C. latent heat of vaporization | D. surface tension |
-
14. A student focused a slide of an onion epidermal cell on pre-estimated width of 2 millimeter field of view under a microscope. What will be the width of each individual cell if you count 8 cells that fit to the field of view during your observation? (1)
- | | |
|------------------|--------------------|
| A. 16 millimeter | B. 0.40 millimeter |
| C. 4 millimeter | D. 0.25 millimeter |
-
15. Which one of the following statements is correct about the differences between eukaryotic and prokaryotic cells? Prokaryotic cells have (1)
- A. small ribosomes whereas eukaryotic cells have large ribosomes
 - B. membrane bound organelles whereas eukaryotic cells don't have it
 - C. cell wall whereas eukaryotic cells don't have it
 - D. division of labor whereas eukaryotic cells don't have it

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- 16.** Which biological tool is correctly matched with its function? (1)
- A. Burette - to measure mass of a substance
 - B. Petri dishes - to culture microorganisms
 - C. Centrifuges - to collect insects
 - D. Theodolite - to measure the rate of water flow
-
- 17.** A student centrifuge a solution containing plant material in order to separate them based on their mass. After 10 minutes of centrifugation some of the components move to the bottom (A), others to the middle layer (B) and the remaining once to the top layer (C). Which alternative shows the mass of the components in decreasing order? (1)
- A. $C > A > B$
 - B. $A > B > C$
 - C. $A = B = C$
 - D. $B > A > C$
-
- 18.** Which one of the following statements shows the future promise of biological sciences? (1)
- A. Using stem cells to repair damaged organs and grow new organ
 - B. Finding alternative living place for human population
 - C. Finding the adaptation mechanism of plant in new condition
 - D. Advising the government on how to conserve environment
-
- 19.** During oxidative phosphorylation, ATP is produced by combining ADP and inorganic phosphate using energy released from (1)
- A. dehydrogenation of pyruvate to oxaloacetate.
 - B. transport of electrons from complex IV to complex I.
 - C. decarboxylation of citrate from NADH/FADH₂ to O₂.
 - D. transfer of electrons from NADH/FADH₂ to O₂.
-
- 20.** Which one of the following comparisons between substrate-level and oxidative phosphorylation is correct? (1)
- During substrate-level phosphorylation
- A. phosphorylation is made in the cytosol, but in oxidative phosphorylation it is made in the inner membrane of mitochondria.
 - B. 34 ATP molecules are produced per glucose molecule, but in oxidative phosphorylation ATP molecules are produced per glucose molecule.
 - C. protons are pumped into the inter-membrane space, but in oxidative phosphorylation they are pumped into the cytosol.
 - D. oxygen is the final electron acceptor, but in oxidative phosphorylation NAD is the electron acceptor.
-
- 21.** Which one of the following diseases and their mode of transmission is correctly matched? (1)
- A. Candidiasis - droplet infection
 - B. Salmonellosis - contaminated food
 - C. Chicken pox - fecal-oral-route
 - D. Common cold - direct contact
-
- 22.** The view of biochemical origin of life hypothesized that the first life-forms appeared in warm primitive ocean and were heterotrophic. Which one of the following statements can support this idea? (1)
- A. The appearance of dust of unknown source with microbial spores that can feed on organic molecules.
 - B. Life is the inherent property of the universe which permits unlimited existence through usage of resource in it.
 - C. The formation of abundant organic molecules due to radiations and the absence of oxygen led to the primitive living organisms.
 - D. Resistant spores from meteorites reached to the earth and used organic molecules to survive in the environment.

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- 23.** Which statement shows the correct comparison between the Lamarck's and Darwin's theory of evolution? (1)
- A. Darwin recognized the origin of species was through mutation while Lamarck believed it was through effect of the environment.
 - B. Lamarck believed in the development of new traits due to variation while Darwin believed in their occurrence due to use and disuse of organs.
 - C. Lamarck believed the gradual occurrence of changes while Darwin believed in the immediate change of species.
 - D. Darwin claimed that variation is heritable while Lamarck considers variation as acquired traits.
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- 24.** Identify the correct mechanism of bacterial reproduction by conjugation. (1)
- A. Mitotic spindles are produced to separate bacterial DNA into two equal parts.
 - B. The recipient bacteria use their pili to pull the donor bacteria during plasmid transfer.
 - C. Bacteria take up genetic materials of dead organisms and integrate into their DNA.
 - D. A pair of bacteria come together and exchanges their plasmids.
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- 25.** Which of the following features is NOT the characteristic feature of fungi? (1)
- A. they reproduce asexually through spore production
 - B. have haploid spores and multinucleate hyphae
 - C. are often autotrophic, parasitic or saprophytic
 - D. exhibit apical growth of networked hyphae
-
- 26.** What is the argument behind the need of appropriate host cell for viruses to replicate themselves? Because (1)
- A. viruses have cell surface antigen specific to a given cell type
 - B. are obligate saprophyte that can only replicate inside a cell
 - C. lack metabolic capacity to produce energy outside cells
 - D. lack of double strand DNA to duplicate themselves
-
- 27.** What is the basic difference between anabolism and catabolism? (1)
- A. Catabolism is condensation while anabolism is hydrolysis reaction
 - B. Anabolism is related to respiration while catabolism is to photosynthesis
 - C. Anabolism is energy absorbing while catabolism is energy releasing process
 - D. Catabolism is a constructive while anabolism is a destructive reaction
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- 28.** What is the main function of the dermis of the skin? (1)
- A. Prevention of the entry of pathogens
 - B. Production of skin pigments
 - C. Synthesis of hair and nail proteins
 - D. Temperature regulation in homeostasis
-
- 29.** A man observed elementary school students spinning around, and realized that none of them could properly walk upright after sudden interruption of the spinning. Which one of the following is the correct reason for this condition? (1)
- A. The sensory cells were unable to detect stimulus after the spinning was quitted.
 - B. The fluid in the semi secular canal stopped immediately when the body stopped spinning.
 - C. The fluid in the semicircular canal kept moving after the spinning was interrupted.
 - D. The nerve impulse was transmitted after the spinning was stopped.

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- 30.** Which of the following is the correct description of hormones? They are chemical messengers that (1)
- A. transmit message faster than the nerve system.
 - B. regulates many physiological process.
 - C. targets only the site of secretion.
 - D. are produced in glands that contain ducts.
-
- 31.** Homeostasis is a mechanism of (1)
- A. adjusting the cell environment relative to body condition
 - B. keeping the internal body environment constant
 - C. keeping the external environment stable
 - D. protecting the body from pathogens
-
- 32.** Which of the following is true concerning the evidences of evolution? (1)
- A. Organisms which live in the same area have similar evolutionary history.
 - B. Organisms which have little genetic relationship show similar behavior.
 - C. Species that are closely related share the most similar DNA sequences.
 - D. Species that live in the same geographic area have the same genetic makeup.
-
- 33.** The type of gene mutation that results in the change of a triplet (GAG) in a normal hemoglobin to another triplet (GTG) in sickle-cell hemoglobin is (1)
- | | |
|-----------------------|----------------------|
| A. insertion mutation | B. deletion mutation |
| C. substitution | D. non-disjunction |
-
- 34.** In human evolution, brain size increased over the course of evolution. Which alternative contains the correct series of brain size? (1)
- A. Homo erectus → Homo habilis → Homo neanderthalensis → Homo sapiens
 - B. Homo neanderthalensis → Homo erectus → Homo habilis → Homo sapiens
 - C. Homo erectus → Homo neanderthalensis → Homo habilis → Homo sapiens
 - D. Homo habilis → Homo erectus → Homo neanderthalensis → Homo sapiens
-
- 35.** Which one of the following statements correctly explains the genetic drift? It is a change in allele frequency due to (1)
- A. non-random mating of individuals in a population.
 - B. independent assortment of chromosomes.
 - C. chance based sampling error.
 - D. relocation of species into other areas.
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- 36.** Which one among the following processes occurs during the light independent phase of photosynthesis? (1)
- A. Excitation of electrons from chlorophyll molecules
 - B. Cyclic photophosphorylation of ADP in the thylakoid
 - C. Splitting of water molecules into its components
 - D. Reduction of triose molecules by hydrogen from NADPH

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- 37.** As compared to non-cyclic photophosphorylation, the cyclic photophosphorylation uses the (1)
- A. photosystem I to produce ATP and reduced NADP in the stroma.
 - B. photosystem I as the last electron acceptor.
 - C. hydrolysis of water to replace electrons lost from photosystem II.
 - D. photosystem II as the first electron donor to NADP.
-
- 38.** Which of the following chemical reactions is the process of photosynthesis? (1)
- A. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$
 - B. $C_6H_{12}O_6 + 6CO_2 \rightarrow C_6H_{12}O_6 + 6O_2 + ATP$
 - C. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$
 - D. $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$
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- 39.** Which one of the following statements properly describes the structure of ATP? It is made up of (1)
- A. two adenine bases attached to a hexose sugar and three phosphate groups.
 - B. one adenine base attached to a pentose sugar and three phosphate groups.
 - C. two adenine bases attached to a deoxyribose sugar and two phosphate groups.
 - D. one adenine base attached to a ribose sugar and two phosphate groups.
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- 40.** Which one of the following descriptions about glycolysis is correct? It is a process (1)
- A. that produces a net of 4 ATP through substrate level phosphorylation.
 - B. of producing acetyl CoA from pyruvate by releasing CO_2 .
 - C. of breaking down of a glucose molecule to two pyruvates.
 - D. that takes place in the cytosol with high demand of oxygen.
-
- 41.** Which one of the following is the main function of Bowman's capsule of the human kidneys? (1)
- A. Prevent back flow of urine
 - B. Mineral reabsorption
 - C. Ultrafiltration of blood
 - D. Water preservation
-
- 42.** What is the justification for feeling warm while sweating a lot in a very humid condition? (1)
- A. Closure of the glands in the epidermal layer of the skin
 - B. The lower rate of evaporation of water from the skin surface
 - C. An increase in metabolic rate as temperature increases
 - D. Unequal distribution of ions on membrane of cells
-
- 43.** Among the following statements, one briefly describes the concept of climate change. It is the (1)
- A. radical change of local weather condition that sustain for less than a year.
 - B. sudden change in an average rainfall of a limited area.
 - C. gradual change of atmospheric temperature over a short period of time.
 - D. global change in the temperature, precipitation and wind conditions.
-
- 44.** As a result of human activities, huge amount of carbon dioxide has been accumulated in the atmosphere. What (1)
is the consequence of this phenomenon?
- A. Increased rate of uptake of carbon dioxide by green plants
 - B. Drop in temperature of the earth and death of aquatic organisms
 - C. Preventing certain amount of heat escaping back to the space
 - D. Reduction in the sea level and death of aquatic animals

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- 45.** What was the main objective of the United Nations Framework Convention on Climate Change (UNFCCC)? (1)
- A. Prevention of dangerous man-made interference with the climate system
 - B. Reduction of the monopoly utilization of border crossing rivers to prevent climate change
 - C. Donation of high carbon projects related to climate change by the developed countries
 - D. Prohibition of developing countries from setting up big industries that could affect the climate
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- 46.** The ability of carnivores to survive in an ecosystem is dependent on the (1)
- A. catastrophic factor.
 - B. abiotic factor.
 - C. edaphic factor.
 - D. biotic factor.
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- 47.** This item is based on the following pyramid of biomass. Which alternative contains organisms that can occupy the above pyramid of biomass from the bottom to the top? (1)
- A. Grass, Cricket, Chicken and Lizard
 - B. Grass, Cricket, Lizard and Chicken
 - C. Grass, Chicken, Cricket and Lizard
 - D. Grass, Lizard, Cricket and Chicken
-
- 48.** The two reverse processes that play a significant role in absorbing and releasing carbon dioxide, respectively are (1)
- A. decomposition and respiration.
 - B. combustion and photosynthesis.
 - C. respiration and assimilation.
 - D. photosynthesis and respiration.
-
- 49.** Which one of the following adaptations is important for survival of polar bear in very cold environments? Having (1)
- A. smaller size
 - B. blubbers under the skin
 - C. thick wrinkled ears
 - D. fine silky fur
-
- 50.** Assume your friend has a blood type that has no antigen on his red blood cells but produces antibodies a and b, loses blood due to a car accident. If blood transfusion is necessary to save his life, your friend can receive blood from someone with (1)
- A. blood type B.
 - B. blood type AB.
 - C. blood type O.
 - D. blood type A.
-
- 51.** A student concluded that the stage of meiosis on a prepared slide of plant tissue he observed was Anaphase II. What do you think was the reason to come to this conclusion? Observation of (1)
- A. divided centromere and separated sister chromatids
 - B. reformed nuclear envelope and reappearance of nucleolus
 - C. one chromosome from each homologous pairs at the poles
 - D. crossing over between non-sister chromatids of homologous pairs
-
- 52.** The binomial system of naming developed by Linnaeus comprises two names. Which hierarchical groups are used? (1)
- A. Kingdom and species
 - B. Order and family
 - C. Genus and species
 - D. Kingdom and phylum
-
- 53.** Which one of the following is the feature of kingdom Monera? Its members are (1)
- A. multicellular with cell walls.
 - B. unicellular without separated nucleus.
 - C. mainly autotrophic and saprophytic.
 - D. reproduced by hypha formation.

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- 54.** Which alternative given below is comprised of harmful groups from the kingdom Monera, Protista and Fungi, respectively? (1)
- A. Candida albicans, Mycobacterium tuberculosis and Trypanosoma
 - B. Trypanosoma, Plasmodium falciparum and Trypanosoma
 - C. Haemophilus ducreyi, Trypanosoma cruzi and Tinea pedis
 - D. Trypanosoma, Plasmodium falciparum and Penicillium
-
- 55.** Which one of the following is a feature of Octopi and squid in phylum mollusca? They have (1)
- A. well-developed brain.
 - B. segmented body with head and thorax.
 - C. lung as a breathing structure.
 - D. hard muscular foot with a hard body.
-
- 56.** Which one of the following research activities has been conducted at Armauer Hansen Research Institute? (1)
- A. Improving crop production through genetic engineering
 - B. Conserving the genes of endemic plants and animals
 - C. Assessing the nutritional status of mothers and babies
 - D. Developing drugs for leishmaniasis and tuberculosis
-
- 57.** Which one of the following statements is correct about the process of digestion in the stomach? (1)
- A. Lipase hydrolyzes fat molecules into fatty acids and glycerol.
 - B. Pepsin hydrolyzes protein molecules into amino acids.
 - C. Amylase hydrolyzes starch molecules into maltose.
 - D. Bile hydrolyzes bigger fat molecules into smaller droplets.
-
- 58.** Which one of the following statements is correct about the role of trypsin in digestion? It acts on (1)
- A. maltose in the mouth.
 - B. lipids in the small intestine.
 - C. starch in the stomach.
 - D. protein in the small intestine.
-
- 59.** Which one of the following descriptions about absorption of nutrients in the digestive system is correct? (1)
- A. Amino acid molecules are directly absorbed from small intestine into the blood.
 - B. Glycerol molecules are first absorbed into the hepatic portal vein and then to lacteal.
 - C. Glucose molecules are first absorbed into the lacteal and then enter into the blood.
 - D. Fatty acids molecules are absorbed from small intestine directly into the blood.
-
- 60.** Which one of the following statements is true about female genital mutilation? It (1)
- A. helps to deliver baby easily.
 - B. Gives social acceptance for marriage.
 - C. imposes risk of infection.
 - D. keeps girls pure and honest.
-
- 61.** Assume that you have a friend who is HIV positive and you are familiar about the case. How can you support your friend? (1)
- A. Ignore your friend with precautions
 - B. Help your friend with precautions to protect you from the virus
 - C. Call to other people to protect you from the virus
 - D. Help your friend without any fear of the virus
-
- 62.** Why are root hairs efficient in uptake of water from the soil? (1)
- A. show positive hydrotropism
 - B. anchor the plant to the ground
 - C. keep rigidity of the cell
 - D. have high surface area to volume ratio

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- 63.** What will happen to a plant if the apical dominance is removed through repeated pruning? (1)
- A. plant elongates and bends to one direction
 - B. plant shades its leaves to withstand the condition
 - C. side shoots grow quickly and the plant becomes bushy
 - D. growth of the lateral shoots will be inhibited
-
- 64.** A farmer sowed cotton seeds in a farm land which fulfills the optimum conditions for germination. After a few days, he observed the newly growing seedlings. Which of the following statements is correct about the farmer's observation? The (1)
- A. remaining of the cotyledon below the ground
 - B. growth of plumule without forming a hook
 - C. growth of plumule above the soil surface
 - D. hypogeal germination of the hypocotyl
-
- 65.** Which one of the following properties of enzymes shows their catalytic nature? (1)
- A. Reducing the activation energy
 - B. Being non-specific to substrates
 - C. Being altered in a chemical reaction
 - D. Changing the nature of the product
-
- 66.** If an enzyme is named as "EC 3.4.11.1." in a systematic naming, what does the number "3" represent? (1)
- A. ultra-subclass of an enzyme that is related to the place where they act
 - B. sub-subclass of an enzyme in relation to its protein nature
 - C. sub-class of an enzyme in relation to the source from where it was discovered
 - D. main class of an enzyme which indicates the type of reaction it catalyses
-
- 67.** In the lock-and-key model of enzymatic action, the activation energy is lowered through (1)
- A. straining of the substrate molecules bound to the active site
 - B. provision of alternative pathway for the reaction
 - C. conformational change of substrate molecules
 - D. permanent modification of the active site of the enzyme
-
- 68.** Which one of the following minerals are used as cofactor for the enzyme alcohol dehydrogenase? (1)
- A. Magnesium ions
 - B. Cobalt ions
 - C. Ferrous ions
 - D. Zinc ions
-
- 69.** What mechanism was used by Gregor Mendel to protect the garden peas from self-pollination? He (1)
- A. sterilized the male part of the flower using chemicals
 - B. opened the flower bud before pollen matured
 - C. covered the flower and protects it from pollinating agents
 - D. selected pure dominant traits for breeding
-
- 70.** A student made an experiment to observe the effect of temperature on catalase of mashed potato. After three parallel trials, he obtained the following results. Temperature (°C) Time taken to collect 33 cm³ in seconds Rate
- | Trial | 1 | 2 | 3 | Mean | cm ³ /sec |
|-------|-----|-----|-----|------|----------------------|
| 10 | 54 | 47 | 43 | 48 | 0.69 |
| 20 | 12 | 14 | 16 | 14 | 2.36 |
| 30 | 5 | 5 | 5 | 5 | 6.6 |
| 45 | 14 | 11 | 11 | 12 | 2.75 |
| 50 | 73 | 71 | 57 | 67 | 0.5 |
| 55 | 119 | 109 | 132 | 120 | 0.27 |
- At which temperature was the reaction fastest? (1)
- A. 30°C
 - B. 10°C
 - C. 50°C
 - D. 55°C

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- 71.** Gregor Mendel crossed two pure-breeding traits of pea plants in his experiment and found only one of the traits expressed in the first generation. This shows that the (1)
- A. two traits in the F1 generation were equally expressed
 - B. expressed trait must be determined by the recessive allele
 - C. alleles masked in the F1 generation were lost due to mutation
 - D. expressed trait must be determined by the dominant allele
-
- 72.** Which one of the following statements correctly describes chromosome structure? (1)
- A. Chromosomes are similar to a gene in structure
 - B. The pair of chromosomes is heterozygous in nature
 - C. The DNA is wrapped with ribose sugar to form a chromosome
 - D. Chromosomes are made up of histone bounded DNA
-
- 73.** The breeding mechanism that improves the performance of animals or plants using particular traits is (1)
- A. selective breeding
 - B. self-breeding
 - C. test cross
 - D. combination of traits
-
- 74.** Which one of the following birth control methods is correctly matched with its function? (1)
- A. Physical method - reduces the lining of the uterus
 - B. Hormonal method - avoids early embryo implantation
 - C. Surgical method - prevents the ovum from reaching to the uterus
 - D. Natural method - prevents follicle development and ovulation
-
- 75.** Which one of the following organic molecules has ester bonds in its molecular structure? (1)
- A. Glycogen
 - B. Triglyceride
 - C. Amino acid
 - D. Cellulose
-
- 76.** Which one of the following statement is correct about the structural organization of hemoglobin? It is formed (1)
- A. through the association of two or more polypeptide chains
 - B. when a polypeptide chain folds itself during protein synthesis
 - C. through folding of secondary structure of a polypeptide chain
 - D. by hydrogen bonding between quaternary and tertiary structures
-
- 77.** The presence of large amount of lipid reduces the density of an animal. Which function of triglycerides is related to this fact? (1)
- A. Buoyancy
 - B. Thermal insulation
 - C. Waterproofing
 - D. Respiratory substrate
-
- 78.** A student conducted a food test on three unknown food samples and obtained results as presented in the table below. No Procedures used Colour changes of the food samples after the experiment Food sample 1 Food sample 2 Food sample 3 1 Mixed with Benedict's reagent and heated Blue Red Yellow 2 Mixed with ethanol and poured into water Colourless Colourless Colourless 3 Mixed with Biuret's reagent Mauve Mauve Mauve 4 Iodine solution Brown Brown Brown What do you conclude from the result of the above experiment? (1)
- A. Proteins and reducing sugars are found in all the food samples at equal concentrations
 - B. Lipids and reducing sugars are absent from all the three food samples tested
 - C. Non-reducing sugars are present in all the food samples at different concentrations
 - D. Reducing sugars are found in two of the samples, but not found in all

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- 79.** According to biological scenario, structure is related to function, which one of the following is correctly described? (1)
- A. Alveoli- supported by cartilage for swallowing food
 - B. Trachea - millions of air sacs for diffusion of O₂ and CO₂
 - C. Nasal passage - large surface area with hairs for filtration of air
 - D. Larynx - hair like strictures for mucus production
-
- 80.** As compared to the exhaled air, the inhaled air contains (1)
- A. 16% oxygen, 4% carbon dioxide and 80% nitrogen
 - B. 70% oxygen, 20% carbon dioxide and 80% nitrogen
 - C. 80% oxygen, 20% carbon dioxide and 80% nitrogen
 - D. 21% oxygen, 0.04% carbon dioxide and 80% nitrogen
-
- 81.** How does oxygen is transported in the human body? It is transported (1)
- A. from the red blood cells to actively respiring tissues by facilitated diffusion
 - B. by simple diffusion along concentration gradient from the lungs to the heart
 - C. from the lungs to actively respiring tissue by hemoglobin of RBC
 - D. by the blood plasma from lower concentration to higher concentration
-
- 82.** Which one of the following statements is well justified, why the left ventricle is tough in nature? Because it (1)
- A. returns deoxygenated blood to the heart
 - B. forces oxygenated blood out of the heart
 - C. returns oxygenated blood to the heart
 - D. forces deoxygenated blood out of the heart
-
- 83.** Which one of the following national parks of Ethiopia is declared as the world's heritage site and comprises endemic animals like Walia ibex? (1)
- | | |
|----------------------------------|--------------------------------|
| A. Simien Mountain National Park | B. Yangudi Rassa National Park |
| C. Rift Valley National Park | D. Bale Mountain National Park |
-
- 84.** Which one of the following plant species is endemic to Ethiopia? (1)
- | | |
|-------------------------|------------------------|
| A. Acacia baramuleta | B. Persea papaya |
| C. Eucalyptus globulose | D. Guizotia abyssinica |
-
- 85.** A mother milks her cow every morning and the milk usually forms curd after three days. What is the first step to investigate why this change happens? (1)
- A. Read scientific articles about the reason for curdling of milk
 - B. Take action to prevent the milk from curdling
 - C. Formulate hypothesis about the reason for milk curdling
 - D. Ask a question why the milk forms curd
-
- 86.** A drug efficacy test was conducted under similar set up on two groups of infected mice (A and B), groups A were the control whereas group B were experimental group. A student gave a placebo to the control group and the drug to the experimental group. At the end, the experimental group got better. Which one of the following is true about the result? (1)
- | | |
|-----------------------------|---------------------------------|
| A. The drug has an effect | B. Conclusion cannot be reached |
| C. The experiment was wrong | D. Placebo has better effect |

87. Suppose you did an experiment in a biology laboratory and need to report your work to your biology teacher. Which one of the following steps is correct to report the finding? (1)
- A. Title→Hypothesis→Procedure→Prediction→Conclusion→Result→Evaluation→Acknowledgment
 - B. Title→Procedure→Result→Hypothesis→Prediction→Conclusion→Acknowledgment→Evaluation
 - C. Title→Hypothesis→Prediction→Procedure→Result→Conclusion→Evaluation→Acknowledgment
 - D. Title→Prediction→Procedure→Hypothesis→Evaluation→Acknowledgment→Result→Conclusion
88. Which one of the following arrangement of molecules in the fluid mosaic model of the cell membrane is correct? (1)
- A. Phospholipid molecules are peripherally attached to proteins
 - B. Phospholipids fill the protein like a sandwich to retain its rigidity
 - C. Phospholipids are inserted into the bilayer of protein molecules
 - D. Protein molecules are studded between the phospholipid bilayers
89. What is the role of cholesterol in the cell membrane? It (1)
- A. reduces the fluidity of the cell membrane
 - B. anchors integral proteins to the cytoskeleton
 - C. acts as receptor site for hormones and other signaling molecules
 - D. transports hydrophilic molecules into the cell
90. This item is based on the following table about the function of cellular organelles. No Features Organelles (1)
- | Organelles | No | Features | Yes | No | Yes | No | Yes | No |
|--------------|----|------------------------------------|-----|-----|-----|-----|-----|-----|
| Mitochondria | 1 | The site of protein synthesis | NO | YES | NO | YES | NO | YES |
| Golgi body | 2 | Modifies and distributes molecules | NO | YES | NO | NO | NO | NO |
| Lysosomes | 3 | Site of aerobic respiration | YES | NO | NO | NO | NO | NO |
| Ribosomes | 4 | Digest foreign cells | NO | NO | YES | YES | YES | YES |
- Which numbers in the above table show the correct function of the given organelles?
- A. 2 and 3
 - B. 2 and 4
 - C. 1 and 4
 - D. 1 and 3
91. Potato slices of equal weighed were immersed in four different concentrations of salt solutions. The solutions were 1.5, 1.0, 0.5, and 0.1 molar. After a day, the potato slices were reweighed. Which potato slice weights the least? (1)
- A. 1.5 molar
 - B. 0.1 molar
 - C. 0.5 molar
 - D. 1.0 molar
92. Suppose you took a container and divided it into two halves by a membrane which is permeable to water and divided the two halves as "A" and "B". Finally, you added hypertonic solution in "A" and hypotonic solution in "B". What will be the expected result of this experiment? The water level of (1)
- A. "B" decrease
 - B. "A" remains constant
 - C. both "A" and "B" rise up
 - D. "B" rise up

— END OF EXAMINATION —